Java : Real-time Stock Price Monitoring System

Hard

Overview

Implement a real-time stock price monitoring system that updates and logs multiple stock prices every second. The prices of the stocks fluctuate based on a predefined percentage increase every second. Goal The system should handle the following for multiple stocks: Formula: The formula used for updating the price of stock: Note: The price is rounded to 2 decimal places when logged, but no rounding is applied when updating the prices. Example Input: Explanation: Output: Explanation: Function Description Complete the function monitorStocks in the editor: This method monitors stock prices, updates them for the given instance (second), and logs the result. Function Parameters: Returns List<String>: A list of logs for the current second, each containing the updated stock prices in the specified format. Class Description Complete the class Stock in the editor: It represents a stock with the following properties: Constructor: Initializes a stock with its name, starting price, and percentage increase per update. Methods: Constraints The first line contains an integer n, the number of stocks. Each of the next n lines contains three space-separated integers: Finally, a line contains the duration for which the stock updates should be updated and logged. Sample Input 0 Sample Output 0 Explanation In this case, we have two stocks: NFLX and DIS. The initial prices and percentage increases per second are as follows: For each stock, we calculate the updated price each second. NFLX (Starting at $250.0, increasing by 3.0% per second): DIS (Starting at $180.0, increasing by 1.8% per second): Over 4 seconds, the updated stock prices are logged in the required format. Sample Input 1 Sample Output 1 Explanation The system logs the updated stock prices each second for 3 seconds. INPUT FORMAT FOR CUSTOM TESTING SAMPLE CASE 0 SAMPLE CASE 1 Update the price of each stock based on a percentage increment every second.Log the updated stock prices every second for a specified duration in seconds.Ensure thread safety when updating and reading stock prices.Initial stock prices: AAPL = $100, GOOGL = $200, AMZN = $150.AAPL increases by 2% every secondGOOGL increases by 1.5% every secondAMZN increases by 2.5% every secondThe stock prices have to be increased each second for 2 seconds. After 1 second: * AAPL = $(100 + 100 * (0.02)) = $102.00 * GOOGL = $(200 + 200 * (0.015)) = $203.00 * AMZN = $(150 + 150 * (0.025)) = $153.75 * Log: Second 1: AAPL: $102.00 GOOGL: $203.00 AMZN: $153.75 After 2 seconds: * AAPL = $(102 + 102 * (0.02)) = $104.04 * GOOGL = $(203 + 203 * (0.015)) = $206.045 = $206.05 (rounded to two decimal places when logged). * AMZN = $(153.75 + 153.75 * (0.025)) = $157.59375 = $157.59 (rounded to two decimal places when logged). * Log: Second 2: AAPL: $104.04 GOOGL: $206.05 AMZN: $157.59 For each stock, the updatePrice() method is called to update its value. The updated values are rounded to 2 decimal places and logged in the following format: * Second X: StockName1: $UpdatedPrice1 StockName2: $UpdatedPrice2 int instance: The current second (instance number) at which the stock prices are being updated. int numberOfStocks: Total number of stocks being monitored. List<Stock> stocks: A list of Stock objects representing each stock's details. name: The name of the stock, e.g., NFLX, AAPL. price: The current price of the stock. percentageIncrease: The percentage by which the stock price increases after each update. void updatePrice(): Updates the stock price by applying the percentage increase. String toString(): Overriden method that returns a string representation of the stock in the format: StockName: $UpdatedPrice 1 ≤ n (number of stocks) ≤ 10 1 ≤ stock price ≤ 5000 0 ≤ percentage of increase ≤ 100 1 ≤ number of seconds ≤ 10 The stock nameThe initial priceThe percentage increase per secondNFLX starts at $250.0 and increases by 3.0% per second.DIS starts at $180.0 and increases by 1.8% per second. After the 1st Second: Updated Price = $(250.00 + 250 * 0.03) = $257.50After the 2nd Second: Updated Price = $(257.50 + 257.50 * 0.03) = $265.225 = $265.23 (rounded to two decimal places when logged).After the 3rd Second: Updated Price = $(265.225 + 265.225 * 0.03) = $273.18175 = $273.18 (rounded to two decimal places when logged).After the 4th Second: Updated Price = $(273.18175 + 273.18175 * 0.03) = $281.3772025 = $281.38 (rounded to two decimal places when logged).After the 1st Second: Updated Price = $(180.0 + 180.0 * 0.018) = $183.24After the 2nd Second: Updated Price = $(183.24 + 183.24 * 0.018) = $186.53832 = $186.54 (rounded to two decimal places when logged).After the 3rd Second: Updated Price = $(186.53832 + 186.53832 * 0.018) = $189.89600976 = $189.90 (rounded

to two decimal places when logged).After the 4th Second: Updated Price = $(189.89600976 + 189.89600976 * 0.018) = $193.314137936 = $193.31 (rounded to two decimal places when logged).
Java (Advanced)
Coding
100 score
40 mins
15 test cases
Java, Problem Solving, Multithreading, ExecutorService

Question

Implement a real-time stock price monitoring system that updates and logs multiple stock prices every second. The prices of the stocks fluctuate based on a predefined percentage increase every second.

Goal

The system should handle the following for multiple stocks:

1. Update the price of each stock based on a percentage increment every second.
2. Log the updated stock prices every second for a specified duration in seconds.
3. Ensure thread safety when updating and reading stock prices.

Formula:

The formula used for updating the price of stock:

$$\text{UpdatedPrice} = \text{CurrentPrice} + \text{CurrentPrice}*(\frac{\text{percentageIncrease}}{100})$$

Note: The price is rounded to 2 decimal places when logged, but no rounding is applied when updating the prices.

Example

Input:

```
3
AAPL 100.0 2.0
GOOGL 200.0 1.5
AMZN 150.0 2.5
2
```

Explanation:

- Initial stock prices: AAPL = $100, GOOGL = $200, AMZN = $150.
- AAPL increases by 2% every second
- GOOGL increases by 1.5% every second
- AMZN increases by 2.5% every second
- The stock prices have to be increased each second for 2 seconds.

Output:

```
Second 1:
AAPL: $102.00
```

```
GOOGL: $203.00
AMZN: $153.75

Second 2:
AAPL: $104.04
GOOGL: $206.05
AMZN: $157.59
```

Explanation:

- After 1 second:
  - AAPL = $(100 + 100 * (0.02)) = $102.00
  - GOOGL = $(200 + 200 * (0.015)) = $203.00
  - AMZN = $(150 + 150 * (0.025)) = $153.75
  - Log:

```
Second 1:
AAPL: $102.00
GOOGL: $203.00
AMZN: $153.75
```

- After 2 seconds:
  - AAPL = $(102 + 102 * (0.02)) = $104.04
  - GOOGL = $(203 + 203 * (0.015)) = $206.045 = $206.05 (rounded to two decimal places when logged).
  - AMZN = $(153.75 + 153.75 * (0.025)) = $157.59375 = $157.59 (rounded to two decimal places when logged).
  - Log:

```
Second 2:
AAPL: $104.04
GOOGL: $206.05
AMZN: $157.59
```

Function Description

Complete the function *monitorStocks* in the editor: This method monitors stock prices, updates them for the given instance (second), and logs the result.

- For each stock, the *updatePrice()* method is called to update its value.
- The updated values are rounded to 2 decimal places and logged in the following format:

```
Second X:
StockName1: $UpdatedPrice1
StockName2: $UpdatedPrice2
```

•

Function Parameters:

- *int instance*: The current second (instance number) at which the stock prices are being updated.
- *int numberOfStocks*: Total number of stocks being monitored.
- *List<Stock> stocks*: A list of *Stock* objects representing each stock's details.

Returns

List<String>: A list of logs for the current second, each containing the updated stock prices in the specified format.

Class Description

Complete the class *Stock* in the editor: It represents a stock with the following properties:

- *name*: The name of the stock, e.g., NFLX, AAPL.
- *price*: The current price of the stock.
- *percentageIncrease*: The percentage by which the stock price increases after each update.

Constructor:

Initializes a stock with its name, starting price, and percentage increase per update.

Methods:

1. *void updatePrice():* Updates the stock price by applying the percentage increase.
2. *String toString():* Overriden method that returns a string representation of the stock in the format:

```
StockName: $UpdatedPrice
```

Constraints

- $1 \leq n$ (number of stocks) ≤ 10
- $1 \leq$ *stock price* ≤ 5000
- $0 \leq$ *percentage of increase* ≤ 100
- $1 \leq$ *number of seconds* ≤ 10

The first line contains an integer *n*, the number of stocks.
Each of the next n lines contains three space-separated integers:

1. The stock name
2. The initial price
3. The percentage increase per second

Finally, a line contains the duration for which the stock updates should be updated and logged.

Sample Input 0

```
STDIN          Function
-----          --------
```

```
2               n = 2
NFLX 250.0 3.0   name = "NFLX", price = 250.0, percentageIncrease = 3.0
DIS 180.0 1.8    name = "DIS", price = 180.0, percentageIncrease = 1.8
4               number of seconds = 4
```

Sample Output 0

```
Second 1:
NFLX: $257.50
DIS: $183.24

Second 2:
NFLX: $265.23
DIS: $186.54

Second 3:
NFLX: $273.18
DIS: $189.90

Second 4:
NFLX: $281.38
DIS: $193.31
```

Explanation

In this case, we have two stocks: NFLX and DIS. The initial prices and percentage increases per second are as follows:

- NFLX starts at $250.0 and increases by 3.0% per second.
- DIS starts at $180.0 and increases by 1.8% per second.

For each stock, we calculate the updated price each second.

NFLX (Starting at $250.0, increasing by 3.0% per second):

- After the 1st Second: Updated Price = $(250.00 + 250 * 0.03) = $257.50
- After the 2nd Second: Updated Price = $(257.50 + 257.50 * 0.03) = $265.225 = $265.23 (rounded to two decimal places when logged).
- After the 3rd Second: Updated Price = $(265.225 + 265.225 * 0.03) = $273.18175 = $273.18 (rounded to two decimal places when logged).
- After the 4th Second: Updated Price = $(273.18175 + 273.18175 * 0.03) = $281.3772025 = $281.38 (rounded to two decimal places when logged).

DIS (Starting at $180.0, increasing by 1.8% per second):

- After the 1st Second: Updated Price = $(180.0 + 180.0 * 0.018) = $183.24
- After the 2nd Second: Updated Price = $(183.24 + 183.24 * 0.018) = $186.53832 = $186.54 (rounded to two decimal places when logged).
- After the 3rd Second: Updated Price = $(186.53832 + 186.53832 * 0.018) = $189.89600976 = $189.90 (rounded to two decimal places when logged).

- After the 4th Second: Updated Price = $(189.89600976 + 189.89600976 * 0.018) = $193.314137936 = $193.31 (rounded to two decimal places when logged).

Over 4 seconds, the updated stock prices are logged in the required format.

Sample Input 1

```
3
MSFT 300.0 1.0
TSLA 400.0 2.0
GOOGL 500.0 1.5
3
```

Sample Output 1

```
Second 1:
MSFT: $303.00
TSLA: $408.00
GOOGL: $507.50

Second 2:
MSFT: $306.03
TSLA: $416.16
GOOGL: $515.11

Second 3:
MSFT: $309.09
TSLA: $424.48
GOOGL: $522.84
```

Explanation

The system logs the updated stock prices each second for 3 seconds.